



OSP Networks Standard

Standard

STANDARD: OSP NETWORKS STANDARD

EFFECTIVE DATE:

NEXT REVIEW DATE:

DESIGNATION	NAME & SURNAME	DATE	SIGNATURE
Managing Executive	Danvig De Bruyn		
Systems Administrator	Dominique Crossman		
Head of Distribution	Godfrey Krige		
Head of Standards and Systems	Ignatius Visser		
Area Manager	Jaco Freysen		
Manager of POP and R&D	Jared Honey		
Area Manager	Johan Small		
Data & Systems	Keagan Stokoe		
CTO	Magnus Rademeyer		
Consultant	Stephanus van Deventer		

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LIST OF ACRONYMS

ODN	Optical distribution network
OFN	Optical Feeder Network
PON	Passive optical network
FTS	First Tier Splitter
STS	Second Tier Splitter
ODF	Optical distribution frame
NAP	Network access point
PFN	Primary Feeder Network
SFN	Secondary Feeder Network
DFN	Distribution Feeder Network
POP	Point of Presence

GLOSSARY

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FOREWORD

In the dynamic landscape of modern technology, the efficient and secure functioning of networks is imperative for the seamless flow of information and the success of businesses and organisations. The Fibertime™ networks team recognizes the significance of adhering to established standards to ensure the reliability, interoperability, and security of our network infrastructure. As a team the Fibertime™ networks will improve on existing standards.

DESCRIPTION

This OSP Network Standards Document serves as a comprehensive guide outlining the OSP build in a fiberoptic network as designed on the principles set out by the Fibertime™ team. It is a testament to our commitment to excellence, aiming to create a robust, scalable, and future-proof network environment.

SCOPE

Building the Fibertime™ OSP network will not be limited to only the GPON network that is driven by Fibertime™ but also expand on the backhaul portion of the Fibertime™ network in cases where the 3rd party is out of initial reach. This document will expand on three major sections namely.

- Civil Environment
- Optical Environment
- 3rd party Environment

Please refer to below reference picture to determine where you are in the network.

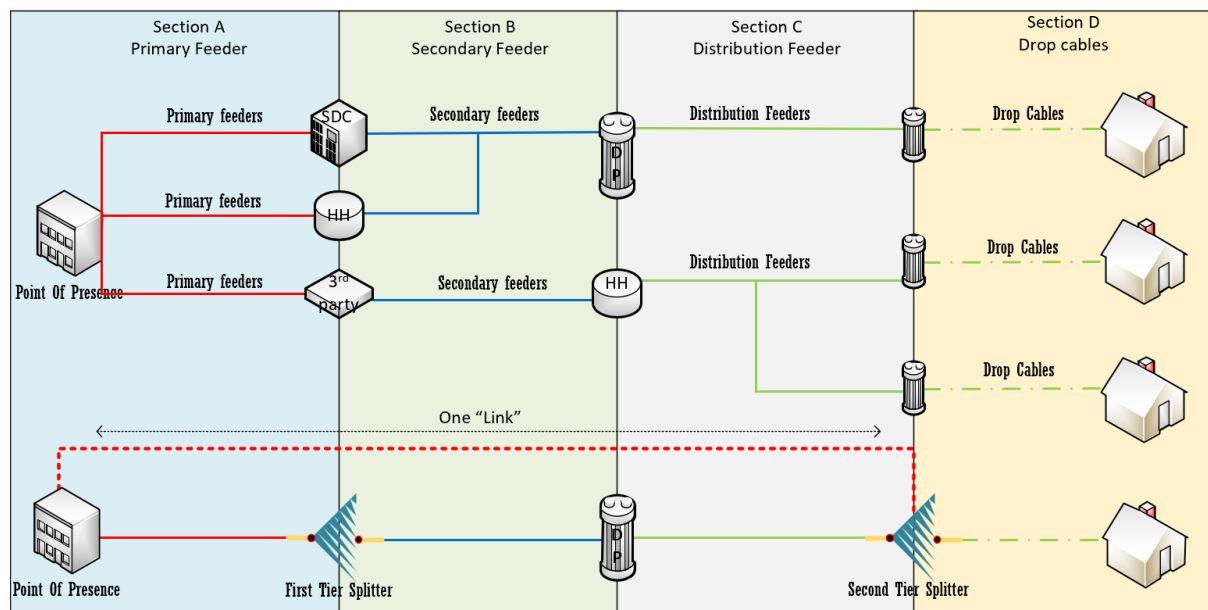


Figure 1 - Network Layout Drawing

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GOVERNANCE

Area Managers: The area managers will be directly responsible for the implementation of the standard set out in this document. This team will also measure the outcomes of such set out standards.

Standards and Planning Manager: This manager is responsible for the implementation of standards and systems throughout the functional streams of the Growth Division in Fibertime

Head of Standards and Systems: This manager is responsible for the implementation of standards and systems throughout the functional streams of the Growth Division in Fibertime™.

Head of Distribution: The head of distribution will report direct changes to the standards and planning manager to improve the ease of install.

POP and Transmission Manager: The POP manager will be responsible to engage with the Backhaul providers and to engage with all the above parties regarding network changes in the POP that will impact the OSP build in any way.

TRAINING

All contractors will be required to obtain training from a certified institute to ensure compliance and competence when working on Fibertime™ sites. This will be a requirement for both Civil and Optical contractors. All team leaders and managers will need a certificate of completion of the following courses.

Training providers:

- Triple Play – Splicing and Testing.
- Terafibre – OSP work install and site management.

Site managers:

Certified Site Management Specialist (SMS) – Terafibre
Outside Plant (OSP) Project Specialist

Team Leaders:

Certified Shared Pole Route Specialist (SPRS)

Splicers:

Certified Fiber Optic Technician (CFOT) Training

Testing:

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Fiber Testing Specialist Course (FTS)

HEALTH AND SAFETY

All contractors will need to comply to the OHSA Act 85 of 1993. This will require that they go through health and safety vetting process before work can commence on-site. They will need a master safety file including, but not limited to, the following:

MASTER SAFETY FILE

1. PPE Issue Register
2. Monthly Health & Safety Inspection
3. Ladder Inspection Register
4. Portable Electrical Equipment Register
5. Hand Tool Inspection Register
6. First Aid Inspection Register
7. Fire Equipment Inspection Register
8. Company Organogram 9. Company Insurance Cover
9. Letter of Good Standing
10. Health & Safety Policies
11. Emergency Procedure with Emergency Contacts
12. Task Specific Risk Assessment
13. Safe Work Procedures
14. Health & Safety Plan
15. Employee Name List with Designations
16. Employee ID Copies
17. Health & Safety Induction 19. Medical Examinations of Fitness
18. HSE Legal Appointments 21. Competence Training Certificates
19. MSDS's Relevant to Scope of Work
20. Incident Reporting Forms (W.Cl.2 & Annexure 1, Incident Register, First Aid Register)
21. Copy of the Occupational Health & Safety Act 85 of 1993
22. Copy of the Construction Regulations 2014 (Government Gazette)
23. TMP -Traffic Management Plan
24. Legal liability
25. DSTI - Daily Safety Task Instruction
26. Project Management Plan.

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The OSP Environment

This environment consists of multiple components and material. We will start a detailed standard of how the network is set out from the POP to the customer going through all the factors in detail, not supplying specifications but guiding standards on implementation.

POP SITE ENVIRONMENTS

The POP site is the heart of our network, meaning it a vital component and without proper care, it will cause major disruptions.

POP Site Civil Environment

The POP site will be the central operation point of the OSP Network.

- The collocated point will be trenched infrastructure.
- Two manholes must be place outside the POP to provide a collocated point for all the optical cables to enter the POP.
- These Manholes will be the first two manholes numbered and named in the area.
Example: “KYA.MH.A001” and “KYA.MH.A002”
- Manholes will be fitted with slack brackets and emergency mounding points for dome joint installation.
- 2x 110mm sleeves will be installed from both the manholes into the POP site with a 7-way 14/10 installed in one of the two in both manholes and a 2-way 14/10 in only manhole “KYA.MH.A001”. When redundancy is in place, both 110mm sleeves from both manholes will be occupied, one with Feeder cables 7-way and the other with the backhaul 2-way.
- The manholes used for this installation will be locked and access controlled.
- The POP site will be installed on a Plinth or pedestal design signed off by a structural engineer.
- All entries into the POP will be sealed with expanding foam and neatened on the inside.
- Entry Point into the POP will be planned and installed into the plinth or buried and encased where a pedestal design is applied.
- The power cable will enter the POP is a 75mm Nextube and will be encased into the POP with no visible sign of the power cable entering (Bottom entry).
- Ducts entering the POP site must be neatly separated and cables tied to the cable tray one by one next to each other.
- Duct Cut-back will be implemented inside the manhole as 1.5 m total duct length with an outer sheathing of 150 mm.

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- Refer to labelling spec for manhole labels.

POP Site Optical Environment

- The optical fiber cabling for the OSP will be collocated in two manholes in front of the POP site and entered in a 50% / 50% ratio into the POP.
- Both Manholes in front of the POP must be supplied with 50m Slack on the primary feeder and Backhaul cable with 15m for maintenance purposes.
- Slack to be installed on a slack bracket and neatly tied back.
- All cabling entering the POP site will be labelled according to the set-out labelling and naming convention by Fibertime™ networks.
- Optical cable exiting the duct will be neatly tied down to the cable tray.
- ISP Optical standards will supply all the details of termination inside the POP site.

POP Site 3rd party environment

- The Optical cable and route shall be trenched underground only.
- Backhaul fiber cabling will be floated separately in the 2way 14/10 duct entering the POP.
- Backhaul fiber will be kept separate from the feeder cables as far as possible.
- Backhaul cables will be marked in the Manholes entering the POP.
- Backhaul Fibre will NEVER be presented in a dome joint with feeder cables outside the POP, it may be spliced in case of emergency but will be installed in a separate dome.
- When entering the POP Backhaul cable and ducting will be marked as high risk.
- Optical cable exiting the duct will be neatly tied down to the cable tray.
- ISP Optical standards will supply all the details of termination inside the POP site.

PRIMARY FEEDER ENVIRONMENT

The Primary feeder is defined as the section between the POP site and the First Tier Splitter Location. This section of the network is not subject to any splitting and carries signal over longer general distances.

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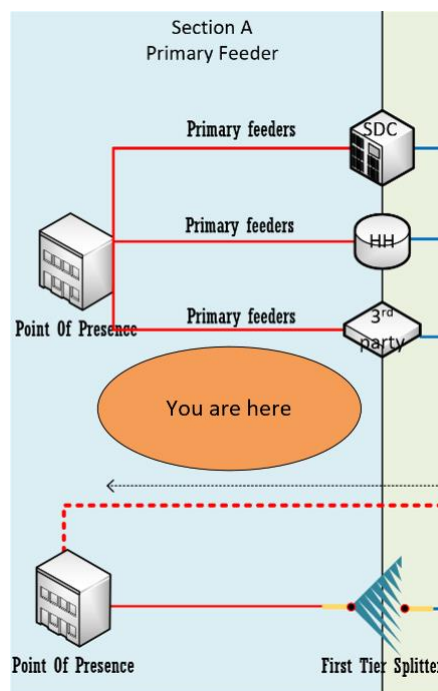


Figure 2 - Primary Feeder

PRIMARY FEEDER CIVIL ENVIRONMENT

- Manholes on this section will conform to the civil specification for installation, cut-back and labelling purposes.
- Manholes on this route might need to comply to monitoring but not access control.
- All poles will be installed as set out in the civil specification.
- Stays and struts must be installed as per the civil specification.
- Pole dressing will conform to on-site example set out in the training process.
- All components of the pole and dressing will be labelled according to the Labelling convention.
- This section will conform to all conditions set out by the wayleave and municipal approvals.

PRIMARY FEEDER OPTICAL ENVIRONMENT

- All Fibres in this section will be tested
- No splitting will be allowed before the first-tier splitter
- No drop cables will be allowed in this section.
- Slack must be allowed on each pole for maintenance purposes.

Table 1 - Slack Summary

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Amount of cable slack per pole	
Long-haul Pole lengths	Slack
CCA H4 SANS 754 (5.4m)	15.8m
CCA H4 SANS 754 (7.2m)	19,4m
CCA H4 SANS 754 (9.0m)	23,0m

- Slack brackets will form part of the civil pole dressing and will be placed in all the locations that breakouts to the secondary feeder cable.
- There will be no back feeding accepted in any part of the network, in other words the fibre itself will be splices from the POP to the customer, a live fibre “link” will not pass a location and then be spliced back to such a location.
- No “spur” cables will be allowed on this section of the network, the primary feeder cables objective is to run past all the secondary feeder point in the network where the first tier splitter is allocated.

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SECONDARY FEEDER ENVIRONMENT

The secondary Feeder environment starts at the end of the primary feeder and ends at the point where it is spliced to a cable size below 12F. This section of the network will contain splicing to the output legs of the splitters, and midspans to route fibres to the closest distribution cables.

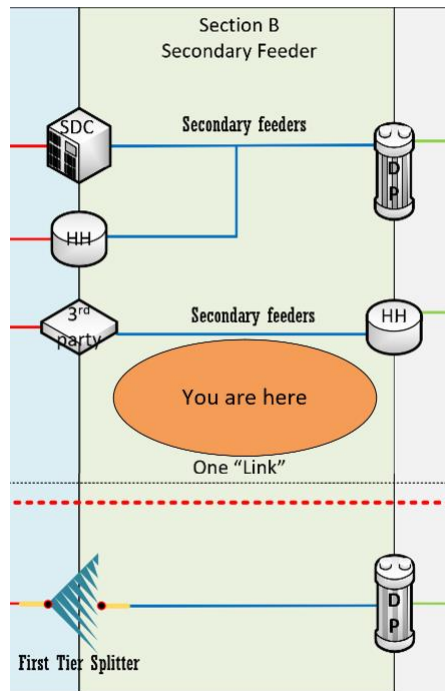


Figure 3 - Secondary Feeder

Secondary Civil Environment

- Manholes on this section will conform to the civil specification for installation, cut-back and labelling purposes.
- Manholes on this route might need to comply to monitoring but not access control.
- All poles will be installed as set out in the civil specification.
- Stays and struts must be installed as per the civil specification.
- Pole dressing will conform to on-site example set out in the training process.
- All components of the pole and dressing will be labelled according to the Labelling convention.
- This section will conform to all conditions set out by the wayleave and municipal approvals.

Secondary Feeder Optical Environment

- Fibre Cables in this section will not require "post installation" testing
- No Splitting will be allowed between the first tier and second tier splitting locations.

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- Drop cables will be allowed in this section as per the design standards document.
- Slack must be allowed on each pole for maintenance purposes.

Table 2 - Slack Summary

Amount of cable slack per pole	
Long-haul Pole lengths	Slack
CCA H4 SANS 754 (5.4m)	15.8m
CCA H4 SANS 754 (7.2m)	19,4m
CCA H4 SANS 754 (9.0m)	23,0m

- Slack brackets will form part of the civil pole dressing and will be placed in all the locations that breakouts to the secondary feeder cable.
- There will be no back feeding accepted in any part of the network, in other words the fibre itself will be spliced from the POP to the customer, a live fibre “link” will not pass a location and then be spliced back to such a location.
- In this section of the network “Spur” links will be allowed to make sure that all sections of the distribution part of the network can be reached with a “live fibre”.
- Splicing from the secondary feeder network to the distribution network will have splitting involved to accommodate drops along the route.
- The SFN will have mainly Mini ADSS cables installed to reduce the need for stays and struts and also the add to speed of deployment.

DISTRIBUTION CABLE ENVIRONMENT

The distribution cable environment will be demarcated as any cable with size lower than a 24F cable down to a 2F cable. This section of the network will be used to effectively expand into all the areas of the targeted market.

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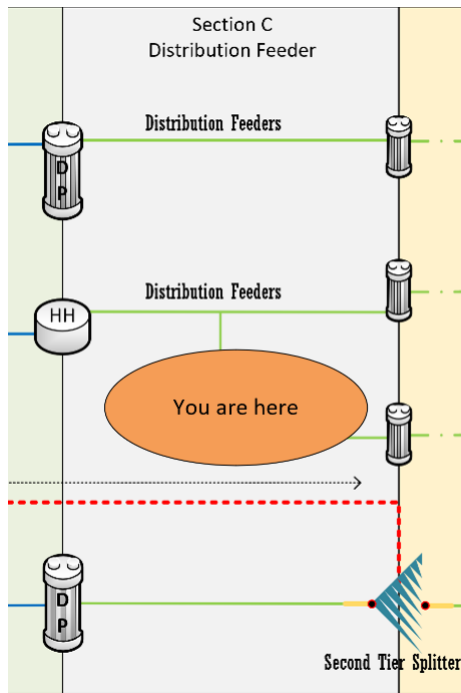


Figure 4 - Distribution Feeder

Distribution Cable Civil Environment

- Manholes on this section will conform to the civil specification for installation, cut-back and labelling purposes.
- Manholes on this route might need to comply to monitoring but not access control.
- All poles will be installed as set out in the civil specification.
- Stays and struts must be installed as per the civil specification.
- Pole dressing will conform to on-site example set out in the training process.
- All components of the pole and dressing will be labelled according to the Labelling convention.
- This section will conform to all conditions set out by the wayleave and municipal approvals.

Distribution Cable Optical Environment

- Fibre Cables in this section will not require “post installation” testing
- No Splitting will be allowed between the first tier and second tier splitting locations.
- Drop cables will be allowed in this section as per the design standards document.
- Slack must be allowed on each pole for maintenance purposes.
-

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Table 3 - Slack Summary

Amount of cable slack per pole	
Long-haul Pole lengths	Slack
CCA H4 SANS 754 (5.4m)	15.8m
CCA H4 SANS 754 (7.2m)	19,4m
CCA H4 SANS 754 (9.0m)	23,0m

- Slack brackets will form part of the civil pole dressing and will be placed in all the locations that breakouts to the secondary feeder cable.
- There will be no back feeding accepted in any part of the network, in other words the fibre itself will be spliced from the POP to the customer, a live fibre “link” will not pass a location and then be spliced back to such a location.
- In this section of the network “Spur” links will be allowed to make sure that all sections of the distribution part of the network can be reached with a “live fibre”.
- This section of the Network will house the secondary splitter
- The DFN will have mainly Mini ADSS cables installed to reduce the need for stays and struts and also the add to speed of deployment.

CUSTOMER PREMISES

The customer premises is the section containing the drop cable from the STS to the actual unit where the ONT will be located. This is the end point of the network and will be the moment of truth once connected, if there are any mistakes in the previous sections this section will carry the burden.

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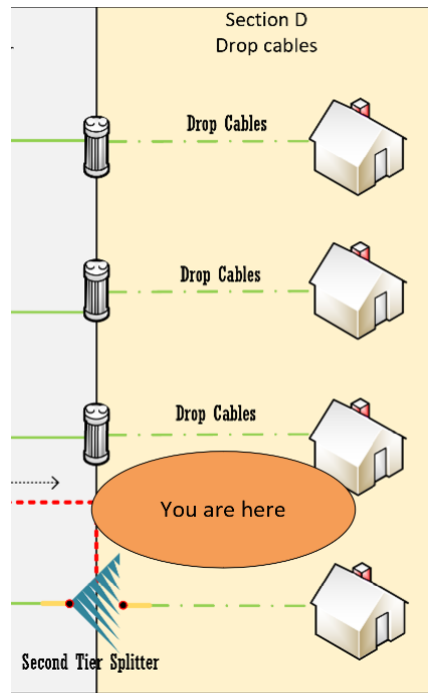


Figure 5 - Customer Premises

Customer Premises Civil Environment

- This section will conform to all conditions set out by the wayleave and municipal approvals.
- The installation from the STS dome enclosure to the premises will comply to the installation standards set out by Fibertime™.
- All entries into the premises will be sealed properly to avoid damage to the customer premises.
- This section will only commence after the approval from the customer has been granted.

Customer Premises Optical Environment

- Slack on the drop cable will be stored on the STS pole in a slack bracket the approved list of material supplied by Fibertime™
- Service will be active after the install is complete.
- All testing requirements set out in the testing specification will be required from Fibertime™ from the STS to the POP for “link” completion.

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VERSION HISTORY

Table 4: Version History

<u>VERSION</u>	<u>FILE NAME</u>	<u>EFFECTIVE DATE</u>	<u>DESCRIPTION OF CHANGE</u>

Source: Fibertime™ (2024)

Table 5: Relevant and Applicable Documents

Document No.	Description
ITU-T G.984.1	GPON General Characteristics Specification
ITU-T G.984.3	GPON Transmission Convergence Layer Specification
FOA	OSP Fiber Optics Civil Works Guide
OHSA 85 of 1993	Occupational Health & Safety Act 85 of 1993
Fibertime™ ALOM	Approved List of Material
Fibertime – Labelling standard	
Fibertime – Naming Convention Specification	

Source: Fibertime™ (2024)